

Job Title: Internship – Research & Development (Mechanical)

- **Department:** Product Development / Research & Development (Mechanical)
- **Target Audience:** Final Year B.Tech / M.Tech (Mechanical / Automotive / Aerospace Engineering)

About the Role

We are seeking passionate, highly motivated, and structurally minded Graduate Engineer Trainees to join our core Research & Development team. As an Intern in R&D, you will sit at the absolute forefront of automotive innovation.

To give you a holistic understanding of our product development cycle, you will initially undergo a structured rotation program across our core engineering domains. Following this cross-functional exposure, you will be permanently aligned to a specific department based on a combination of your personal preference, technical aptitude, and business requirements.

Rotational Domains & Core Functions

As an R&D Intern, you will get hands-on exposure to advanced engineering verticals, including but not limited to:

- **Body-in-White (BIW):** Designing the structural skeleton of the vehicle, focusing on sheet metal design, materials, and welding/joining technologies.
- **Finite Element Analysis (FEA):** Conducting structural, crash, and durability simulations to validate component integrity before physical prototyping.
- **Computational Fluid Dynamics (CFD) & Thermal Management:** Simulating external aerodynamics, under-hood thermal behavior, and HVAC flow optimization.
- **Vehicle Dynamics (Suspension & Steering):** Tuning ride, handling, and steering characteristics, designing hardpoints, linkages, and actuation mechanisms that define the vehicle's driving DNA
- **Powertrain & Transmission Systems:** Developing and packaging next-generation manual, automatic, or electrified drivetrain components.
- **Electrical & Electronics (E&E) Packaging Design:** Strategizing the physical mounting, wiring harness routing, and thermal shielding of complex electronic control units (ECUs).
- **Vehicle Integration:** Managing digital mock-ups (DMU) to ensure zero component interference, tolerance stack-up & BOM management.

Key Responsibilities

- **Design & Development:** Assist in creating 3D CAD models and 2D manufacturing drawings using industry-standard software (e.g., Solidworks, CATIA, Creo, or NX).
- **Simulation & Analysis:** Set up basic boundary conditions for virtual testing, analyze mesh quality, and interpret simulation results alongside senior engineers.
- **Prototyping & Testing:** Cooperate with the testing and validation teams to benchmark physical prototypes against virtual simulation results.
- **Root Cause Analysis:** Participate in solving complex engineering challenges using structured problem-solving methodologies (e.g., DFMEA, Fishbone analysis).
- **Cross-Functional Collaboration:** Coordinate with Sourcing, Manufacturing, and Quality teams to ensure designs are optimized for Manufacturing and Assembly (DFMA).

Candidate Profile & Qualifications

Academic Requirements:

- **Degree:** B.E. / B.Tech / M.Tech in Mechanical Engineering, Automobile Engineering, or Production Engineering.
- **Academic Standing:** Minimum 70% or 7.0 CGPA throughout 10th, 12th, and Graduation (with no active backlogs).

Technical Competencies (What we look for during interviews):

- Strong foundational knowledge of **Strength of Materials, Fluid Mechanics, Thermodynamics, and Kinematics of Machinery**.
- Familiarity with at least one **3D CAD tool** (CATIA, Creo, NX, SolidWorks) or **CAE tool** (Ansys, HyperMesh, Abaqus, Altair).
- Basic understanding of GD&T (Geometric Dimensioning and Tolerancing) and manufacturing processes like casting, forging, and sheet metal forming is a strong plus.
- Exposure to automotive major/minor projects, formula student, or BAJA/SAE events is highly preferred.

Behavioral Skills:

- High learnability and adaptability to transition smoothly between different engineering domains.
- Analytical mindset with a structured approach to problem-solving.
- Excellent verbal and written communication skills to articulate technical ideas clearly.

What We Offer

- A fast-tracked career pathway in core automotive R&D.
- Mentorship from industry-leading subject matter experts (SMEs).

- Hands-on experience with cutting-edge simulation software and state-of-the-art testing labs.